# Estimating a shoebox room from a single room impulse response with unknown playback latency: simulated performance and measured failure of image-source lattice inversion

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## Abstract

When one device plays the excitation and another records it, the first sample of the room impulse response is not the emission time, and the echoes picked from it carry no labels and may include misses and spurious peaks. Every single-response method based on echo arrival times that this search found and read in full assumes either a known emission time or labelled echoes (OpenAlex, arXiv, Crossref, web search and citation tracking, 1979–2026, last searched 2026-10-23). This paper presents a lattice inversion that fits the full image-source lattice of a shoebox room to the unlabelled echoes of one omnidirectional response and estimates the side lengths, the playback latency and the source bearing together; one response fixes only the magnitude of each axis offset, so the bearing is folded into 0–90°. A cross-axis identity between image sources yields closed-form latency hypotheses, candidate lattices are scored by a Poisson point-process likelihood with a clutter term and refined by expectation–maximization, and every fit reports a reliability flag and Laplace standard errors; all settings were frozen before the evaluation. In 500 simulated planar rooms drawn from the population of this study (sides of 3–10 × 3–8 m, source and microphone at least 0.3 m from the walls), 97.8% of the fits came within 5 cm in summed side error, 2.2 percentage points below the variant given the true latency, and the flag marked all 11 failures (failure detection 95% CI 74.1–100%). Given the correct labels, the time-difference least-squares method of Moore et al. [9] succeeded on all 500 of the same responses, so what the lattice inversion adds is that it needs no labels. Success fell to 87% in small rooms and 79% with the microphone near a wall; in three dimensions it was 56%, and 6 of the 118 three-dimensional fits flagged reliable were wrong. Nearly every peak picked in these simulations was a true arrival; at recording signal-to-noise ratios of 10 and 0 dB, with many spurious peaks, success fell to 12% and 0%, and at 0 dB, as well as when inverting with a wrong speed of sound (98 and 97 of 100 fits), the flag still marked wrong fits as reliable. On 540 measured dEchorate responses the method failed every time. The flag rejected every fit, including the 162 that succeeded with known labels, so on measured responses it did not tell right from wrong; it has been validated only within the simulated distribution. An exploratory analysis that was not pre-specified shows that the 120 picked peaks span only about 19 ms with about 40 image arrivals inside; an analysis added after review shows that the frozen scoring model gives wrong lattices a higher likelihood than the fits with known labels, and a diagnosis whose criteria were fixed before its responses were read shows that in single responses of this room, of the first-order echoes only that of the ceiling, when the ceiling

is reflective, matches the calibrated geometry at the centimetre level, the side-wall echoes that fix the horizontal sides cannot be told apart from random times, and neither a matched filter with the direct sound as its template nor envelope peak picking helps; by this diagnosis the failure comes mainly from the echo information a single response of this dataset carries, not only from the peak picker. Applying the lattice inversion to measurements likely needs recordings in which the side-wall echoes can be resolved, or responses from several positions, evaluated on new data.

**Keywords**: room geometry inference, room impulse response, image-source method, echo labeling, emission-time estimation, Poisson point process, reliability assessment

# 1. Introduction

## 1.1 Inferring a room from a single response

The question of whether a shape can be heard goes back to Kac [1], who asked whether the eigenfrequencies of a drum determine its shape. In a shoebox room the specular reflections from the walls are equivalent to a whole lattice of image sources [2], so the arrival time of every echo in an impulse response carries information about the side lengths of the room and the positions of the source and the microphone. Room geometry can serve as prior information for three-dimensional audio rendering and sound reproduction [3], [4], and it is also used for acoustic localization and mapping by robots [5], [6]. Methods that reconstruct a room from echoes mostly use several microphones, several sources or a moving device [7], [8]. Inverting a single room impulse response, with one loudspeaker and one microphone, needs the least equipment: Moore et al. [9] address this problem in planar rectangular rooms, and Yu and Kleijn [10] estimate the source and receiver positions, the room geometry and the reflection coefficients together from a single response in three-dimensional shoebox rooms.

## 1.2 Unknown playback latency and unlabelled echoes

Methods that invert a room from echoes usually take sample 0 of the response as the emission time [10], [11]. Laboratory measurements can meet this assumption: the dEchorate dataset computes the emission time from an additional loopback channel [12]. When playback and recording run on two devices, or when all one has is a response file exported by measurement software, an unknown playback latency separates sample 0 from the emission time. Converted to path length, the latency adds directly to every echo: at 16 kHz, one sample of uncompensated latency is 2.14 cm of path. In a measurement system that synchronizes loudspeakers and phones over Bluetooth, 95% of the synchronization errors were within ±200 μs, which the authors note corresponds to a ranging error of about ±6.8 cm [13].

The second difficulty is that the echoes carry no labels. Peak picking yields only a list of arrival times, without telling which image each one belongs to. Peaks in real responses are also blurred by the loudspeaker characteristics or by several reflections arriving together [12], and the first-order reflection from the wall directly behind the loudspeaker may be too weak to appear in the response [12]. The picked peaks therefore both miss true arrivals and include spurious peaks that correspond to no image. Self-

calibration studies with multiple sensors already treat missing data and outliers as part of the problem [14], [15]. Within the search scope described in Section 1.3, every single-response method read in full assumes away at least one of the two: Moore et al. [9] assume labelled echoes, and Yu and Kleijn [10] and Marković et al. [16] assume a known emission time (Section 2).

### 1.3 Approach and contributions

This paper proposes lattice inversion: the full image-source lattice of a shoebox room is fitted directly to all the echoes picked from a single response, estimating the side length along each axis, the source and microphone positions, the playback latency and the source bearing together, without assigning any echo to a particular wall. The paper makes two contributions. First, an identity between images on different axes gives closed-form latency hypotheses; with unlabelled echoes every triple of echoes is treated as one candidate, and the same identity then sorts the echoes into groups of single-axis images for each axis to generate candidate rooms (Sections 3.2–3.4). This identity is Theorem 1 of Yu and Kleijn [10], and its simplest case is equivalent to the source-to-microphone distance formula of Moore et al. [9]; within the search described in the next paragraph, the only novelty claimed here is its use as a candidate generator for unlabelled echoes. Given the labels, the time-difference least-squares method of Moore et al. also succeeds on every one of the same simulated responses (Section 5.1), so the value of this point lies in not needing labels, not in accuracy. Second, the approach is evaluated with pre-specified simulation experiments with frozen parameters and with a set of real measurements; every fit reports a reliability flag and Laplace standard errors, and the conditions under which it fails are reported as they are: three dimensions, low signal-to-noise ratio, model mismatch and measured responses. Candidate lattices are scored with a Poisson point-process likelihood that has a clutter term and an expected number of peaks $p_{\mathbf{k}}$ for each image; the form is borrowed from multi-target tracking [17] and radar map estimation [18], and this paper uses its Poisson approximation (Section 3.5). In the evaluation scenarios, removing its expected-count term did not change the outcome of any scenario (Section 5.4), and the number of spurious peaks in measured responses far exceeds its clutter setting (Section 5.6), so this paper treats it as a modelling choice and does not list it as a demonstrated contribution. In the evaluation scenarios nearly every picked peak was a true arrival: against the true lattice, the median number of spurious peaks per scenario was 0 in both E1 and E2 (at most 3 and 9), and the median number of missed strong arrivals ($p_{\mathbf{k}}$ at least half of its maximum, that is 0.45, as in Section 3.7) was also 0 (0–4 per scenario in E1, with 67/500 scenarios missing at least one; 0–4 and 10/200 scenarios in E2), whereas at 10 dB and 0 dB the median numbers of spurious peaks were 24 and 37 (an exploratory calculation after review). The results of this paper therefore cannot answer what happens when spurious peaks are numerous (Sections 5.3 and 5.6).

Within the scope of this paper's search (OpenAlex, arXiv, Crossref, web search and citation tracking, covering the literature from 1979 to 2026, last searched on 2026-10-23), the novelty claim covers only geometry inversion based on echo arrival times: of the methods of this kind read in full, none handles a single response, an unknown latency and unlabelled echoes together. The methods of this kind read in full are those of Moore et al. [9], Yu and Kleijn [10], Marković et al. [16], Sprunck et al. [11], Crocco et

al. [15] and Dokmanić et al. [19]; also read in full were the dataset paper of Di Carlo et al. [12] and the paper of Allen and Berkley [2], and the remaining sources were read only in their abstracts or in specified sections. The nearest prior work is Moore et al. [9], which removes the unknown time offset from a single response by time differences but assumes labelled echoes. Marković et al. [16] likewise use only a single response and avoid echo labelling by a global fit, but they measure arrival times from the emission time and need the room shape and the relative positions of source and microphone in advance. Dokmanić et al. [19] likewise use only a single response and need no labels, but they obtain the delays from a known emitted pulse and require all first- and second-order echoes to be heard. Crocco et al. [15] make the onset times optimization variables and need no labels, but they need several microphones and several source positions. Among the single-response methods read only in their abstracts, two have assumptions that cannot be judged from the abstract; see Section 2.1.

### 1.4 Results in brief and organization

In 500 simulated planar rooms, 489 lattice-inversion fits (97.8%) came within 5 cm in summed side error, and the reliability flag marked all 11 failures; in three-dimensional rooms it succeeded in only 56%. On the 540 measured empty-room responses of dEchorate, lattice inversion failed every time; the flag rejected every fit, but it also rejected the fits that succeeded with known labels, so on measured responses it did not tell right from wrong. A diagnosis added after review points to the data themselves: in single responses of this dataset the side-wall echoes cannot be resolved, and the other peak pickers tried did not help (Section 5.7). Section 2 compares the assumptions of existing methods, Section 3 describes lattice inversion, Section 4 sets out the experimental design, Section 5 reports the results, Section 6 discusses the gap between the simulated and the measured results and the limitations of this study, and Section 7 concludes.

## 2. Related work

### 2.1 Inferring rooms from echoes

Methods with several microphones and several sources are the most developed. Dokmanić et al. [8] determine the correspondence between echoes and walls from the properties of Euclidean distance matrices and prove that, under mild conditions, the first-order echoes uniquely determine a convex polyhedral room; arrival times at distributed microphones can be turned into elliptical constraints, with a Hough transform resolving the ambiguity among several walls [7]. Responses from a loudspeaker array to a single microphone can be stacked to estimate image microphones [20], or first-order reflection candidates can be enumerated and candidate rooms then scored by higher-order reflections [3]; scoring by higher-order reflections is close to the approach of this paper, but that method needs a loudspeaker array. Other settings include spherical microphone arrays [21], mobile devices with a co-located loudspeaker and microphone [22], [23], and a phone recording at several places [13]. Drones, ground vehicles and robots carrying microphones can also reconstruct rooms from echoes [24], [25], [26].

A single response carries less information. In planar rectangular rooms, Moore et al. [9] show that the

second-order reflection between the two walls of one axis determines the source offset along that axis, and that this offset is a function of the source-to-microphone distance; Yu and Kleijn [10] use, in three dimensions, the direct sound, the first-order reflections and second-order reflections along a single axis. Dokmanić et al. [19] prove that in a convex polygonal planar room the delays of the first- and second-order echoes in a single room impulse response (RIR) determine the room uniquely (up to rotation and reflection), and they solve the assignment of echoes to walls by searching over permutations with a rank condition (from the full text of their 2011 conference paper). Their assumptions are that the source and the microphone are co-located and both omnidirectional and that the delays are obtained from a known emitted pulse [19]; delay estimation itself lies outside that paper [19]. The room must also let all $K$ first-order and $K(K-1)/2$ second-order echoes of its $K$ walls be heard [19], and the simulations add Gaussian noise only to the delays, with no missed or spurious peaks [19]. This is a precedent for a single response without labels, but the emission time is known, so it does not change the novelty boundary of Section 1.3. Zamaninezhad et al. [27] locate a pair of opposite reflecting surfaces from the resonance frequencies of a single transfer function and, according to their abstract, do not use echo arrival times; Parhizkar et al. [28] localize a microphone in a known room from a single-channel response. Data-driven approaches estimate room dimensions from a single response with convolutional networks [29], or learn the relation between low- and high-order reflections so as to estimate the geometry even when first-order reflections are missing [4]. Neither Zamaninezhad et al. nor Yu and Kleijn [29] invert the geometry from arrival times, so neither falls in the class of methods covered by the novelty claim of Section 1.3; this paper read only the abstracts of these two, and the abstracts do not say how they depend on the emission time. Another line of work casts the recovery of image sources as a sparse problem over Radon measures: Sprunck et al. [30] recover image sources in continuous three-dimensional space, Sprunck et al. [11] recover all the parameters of a room from the responses of a microphone array of known geometry, and Deleforge et al. [31] give analytical properties of shape reconstruction for cuboid rooms.

The geometry that a single response can determine has limits. Even when a solution exists, it is determined only up to symmetries: the parameter classification of Moore et al. [9] shows that the source coordinates can often be determined only up to sign, Yu and Kleijn [10] list, in their three-dimensional setting, a 96-fold degeneracy that sound alone cannot remove, and Kreković et al. [32] fully characterize uniqueness when points and planes are localized from point-to-plane distances. Section 3.8 gives the numbers of equivalent layouts in two and three dimensions and compares them with this 96-fold count.

## 2.2 Emission time and time offsets

The line of work that reverses the image-source method assumes a known emission time; its measurement model assumes an ideal pulse emitted at $t = 0$ [11], [31]. Yu and Kleijn [10] convert arrival times directly into path lengths, which amounts to treating the emission time as known. Studies that estimate time offsets from data rely on several sources and receivers. Time differences of arrival can directly yield the microphone and source positions [33], and self-calibration from times of arrival can tolerate missing data and outliers [14]. With unsynchronized reference transmitters, the receiver positions and the time offsets can be solved together [34]; the array geometry can be recovered when both the emission times and the

receivers' time references are unknown [35]; or the unknown recording-start and emission times can be estimated first to synchronize microphones and sources [36]. For room geometry, Crocco et al. [15] make the onset times optimization variables when both the microphone and the source positions are unknown, but they need several microphones and several source positions.

Within the scope of this paper's search (Section 1.3), of the single-response methods read in full only Moore et al. [9] handle an unknown time offset: the response is written as $h(t-\kappa)$, the unknown $\kappa$ is removed by time differences of arrival, and the source-to-microphone distance is then obtained in closed form from two first-order reflections and the second-order reflection from two adjacent walls [9]. This approach assumes that every arrival is labelled with the image it belongs to and that the direct sound and the required reflections are observable.

### 2.3 Echo labelling and label-free fitting

Echo labelling is the central difficulty of multi-microphone room reconstruction, and dedicated algorithms exist based on maximum independent sets [37], convex hull expansion [38] and Euclidean distance matrices [39]. A study that jointly estimates the arrival times and directions of echoes by expectation–maximization (EM) also notes that labelling remains difficult even once the arrival times are available [40]. Another class of work avoids peak picking and estimates the times and weights of echoes directly in a continuous parameter space [41], [42]; this paper still uses peak picking, and whether that choice has a cost on real measurements is discussed in Section 6.4.

Letting candidate geometries explain the measurement without labelling first also has precedents. Marković et al. [16] use the bidirectional nearest distance between measured and simulated arrivals as the cost for a single response and search with a genetic algorithm, and because the cost is highly nonlinear they have to assume a room class in advance; this cost tolerates spurious peaks but has no model of the detection probability. Tuna et al. [3] score candidate rooms by higher-order reflections; Crocco et al. [15] bypass labelling with a grid search wall by wall; Sprunck et al. [11] find the room axes from the structure of the image-source lattice. Yu and Kleijn [10] use Theorem 1 to separate single-axis from multi-axis reflections and test combinatorial hypotheses about the echoes.

### 2.4 Missed peaks, spurious peaks and reliability

Multi-target tracking describes missed detections and false measurements with Poisson clutter and a detection probability; this is the measurement model of probability hypothesis density (PHD) filtering [17], [43]. Radar map estimation writes the detections as an inhomogeneous Poisson process whose intensity is a fixed clutter level plus an unnormalized Gaussian mixture, and handles the correspondence between measurements and landmarks with variational Bayesian EM [18]; point set registration likewise treats one point set as the centres of a Gaussian mixture and aligns another set to it by EM [44]. Acoustic simultaneous localization and mapping (SLAM) must also handle missed and false direction-of-arrival estimates [6]. The scoring in this paper borrows the Poisson clutter of these models but not the detection probability. Instead, the expected number of peaks $p_{\mathbf{k}}$ of each image describes which arrivals are picked,

which is a Poisson approximation (Section 3.5), and the positions of the mixture components are constrained by the image-source lattice. Estimating by EM the probability that each measurement belongs to each component, rather than enumerating assignments of measurements to targets, is what this paper shares with probabilistic multi-hypothesis tracking (PMHT) [45].

On reliability: among the sources read in full in the search of Section 1.3, the single-response method returns a single configuration with no measure of confidence [10], and the failure rate of the array method can only be counted afterwards [11]; among the sources read only in their abstracts, the nearest thing to a per-fit reliability is a subnetwork that estimates the probability that each wall exists [4]. The simulations of Yu and Kleijn [10] show the failure rate rising from 0.7% to 18.9% as the scattering coefficient increases from 0 to 0.2: single-response inversion fails under model mismatch, and users need to know which run failed. The Laplace approximation, which estimates the standard errors of parameters from the curvature, is an established method [46].

## 2.5 Comparison of assumptions

Table 1 summarizes how the five nearest methods and lattice inversion handle five assumptions.

**Table 1.** Assumptions of existing methods and of lattice inversion

| Method | Microphones/sources | Emission time | Echo labelling | Missed/spurious peaks | Reliability |
|---|---|---|---|---|---|
| Moore et al. [9] | 1/1, planar | Unknown (removed by time differences) | Assumed labelled | Direct sound must be observable; analyses which parameters remain solvable when arrivals are missing; spurious peaks not handled | None |
| Yu and Kleijn [10] | 1/1, three-dimensional | Known (converted directly from arrival times) | Combinatorial hypothesis tests | Robust to erroneous pulses; needs all the required arrival times to be present | None |
| Marković et al. [16] | 1/1, planar; room shape and relative positions of source and microphone known | Known (arrival times compared directly) | Not labelled (global fit) | Bidirectional nearest-arrival distance, tolerates spurious peaks | None |
| Sprunck et al. [11] | Array/1, three-dimensional | Known ($t = 0$) | Axes found from the lattice structure | Merging and thresholds | Failure rate counted afterwards only |

| Method | Microphones/sources | Emission time | Echo labelling | Missed/spurious peaks | Reliability |
|---|---|---|---|---|---|
| Crocco et al. [15] | Several/several positions, all unknown | Estimated (onset times as optimization variables) | Grid search wall by wall, no labels needed | Robust cost function | None |
| Lattice inversion (this paper) | 1/1, planar and three-dimensional | Estimated (closed-form hypotheses plus fitting) | Not labelled | Poisson clutter and expected number of peaks per image | Flag and Laplace standard errors |

**Table 1 notes**: the versions each row is based on. Yu and Kleijn: full text of the arXiv preprint (no published version found), with the "missed/spurious peaks" cell based on its abstract. Marković et al.: full text of the WASPAA (Workshop on Applications of Signal Processing to Audio and Acoustics) 2013 paper. Sprunck et al.: the arXiv version uploaded after publication, checked against the abstract of the published version. Crocco et al.: full text of the published version. The lattice inversion row describes functions only and is not a performance claim; its effect is reported in Section 5.

A single response, an unknown emission time, and unlabelled echoes with both misses and extras: each of these three assumptions has been relaxed by some existing method, but within the scope of this paper's search (Section 1.3), none of the methods read in full relaxes all three together. Dokmanić et al. [19] relax labelling for a single response, but the emission time is known, the source and microphone are co-located, and the echoes must be complete (Section 2.1). Every component of lattice inversion has precedents: the identity and the closed-form latency come from Yu and Kleijn [10] and Moore et al. [9], the Poisson measurement model from tracking and map estimation, and EM and gradual smoothing are established optimization techniques. What this paper examines is how far these components, put together, can go on a single response; the next section describes in turn the latency hypotheses, the bearing, candidate generation, scoring, fitting and reliability.

# 3. Method

## 3.1 Measurement model and notation

This paper uses "shoebox room" for both the planar rectangle and the three-dimensional cuboid room. The observations come from the identified response: its absolute value is taken and divided by the peak value, samples below 0.1% of the peak are set to 0, and peaks are then picked by the rule given at the end of this subsection, with each peak position interpolated to sub-sample precision by a three-point parabola. No envelope, matched filtering or transducer compensation is applied before peak picking; in simulation this is enough to separate the arrivals, but in measured responses it is not (Section 6.4).

Consider a $D$-dimensional shoebox room ($D = 2$ for a planar room, $D = 3$ for a three-dimensional one) whose side along axis $i$ has length $L_i$, with microphone and source coordinates $m_i$ and $s_i$ on that axis.

By the image-source method [2], the image with index $\mathbf{k} \in \mathbb{Z}^D$ is reflected $|k_i|$ times on axis $i$, and its offset relative to the microphone is

$$\Delta_{\mathbf{k},i} = (-1)^{k_i} s_i + 2\lceil k_i/2 \rceil L_i - m_i,$$

its geometric path length is $q_{\mathbf{k}} = \|\Delta_{\mathbf{k}}\|$, and the direct sound has $q_0 = \|\mathbf{s} - \mathbf{m}\|$. The playback latency multiplied by the speed of sound, that is, converted to path length, is denoted $\delta$, and the measured path length is

$$r_{\mathbf{k}} = q_{\mathbf{k}} + \delta.$$

The unknowns are $\theta = (\mathbf{L}, \mathbf{m}, \mathbf{s}, \delta)$. The observations are the peaks picked from the response: starting from the first peak that reaches half of the maximum (taken as the direct sound), peaks above 2% of the direct sound are kept in time order, at most $N$ of them (40 in planar rooms, 120 in three dimensions), with at least 2 samples between neighbouring peaks, and their arrival times are multiplied by the speed of sound to give path lengths $r_1 < r_2 < \cdots < r_N$. These peaks carry no labels; arrivals of some images may be missing among them, and they may contain spurious peaks that correspond to no image. Below, "echoes" always means these picked peaks, and "arrivals" means the predicted or true arrivals of images.

### 3.2 Closed-form latency hypotheses

Take two images $A$ and $B$ whose reflections lie on disjoint axes, for example a side-wall reflection and a floor reflection. Their combination $A{+}B$, which takes the offset of $A$ on the axes where $A$ reflects, the offset of $B$ on the axes where $B$ reflects, and the offset of the direct sound on the remaining axes, is also an image (Fig. 1). Because the squared path length is the sum of the squared offsets along the axes,

$$q_{A+B}^2 + q_0^2 = q_A^2 + q_B^2,$$

which is Theorem 1 of Yu and Kleijn [10] and holds for two-axis reflections of any order. Substituting $q = r - \delta$, the $\delta^2$ terms cancel and the latency has a closed form:

$$\delta = \frac{r_{A+B}^2 + r_0^2 - r_A^2 - r_B^2}{2\,(r_{A+B} + r_0 - r_A - r_B)}.$$

With $d_X = r_X - r_0$ the path difference relative to the direct sound, the same identity gives $q_0 = (d_A^2 + d_B^2 - d_{A+B}^2)/(2(d_{A+B} - d_A - d_B))$, which is exactly the formula with which Moore et al. [9] obtain the source-to-microphone distance from two first-order reflections and the second-order reflection from two adjacent walls; their distance $R_0$ is $r_0 - \delta$. The latency formula itself is therefore not new. The difference is that the formula of Moore et al. is applied to labelled arrivals, whereas here $A$ and $B$ can be images of any order on different axes, and every unlabelled triple of echoes is only a hypothesis.

Hypotheses are generated and screened as follows. Each pair $(A, B)$ among the first 11 reflection echoes, combined with each later echo as $A{+}B$, gives one $\delta$; combinations whose denominator is smaller than 0.2 m are discarded. Each hypothesis is scored by its support among early pairs of echoes: given $\delta$, a pair of echoes $(a, b)$ adds a point to the hypothesis if an echo lies near the predicted position of their combination, $\sqrt{q_a^2 + q_b^2 - q_0^2}$. The top 8 hypotheses that are at least 8 cm apart go on to the next step.

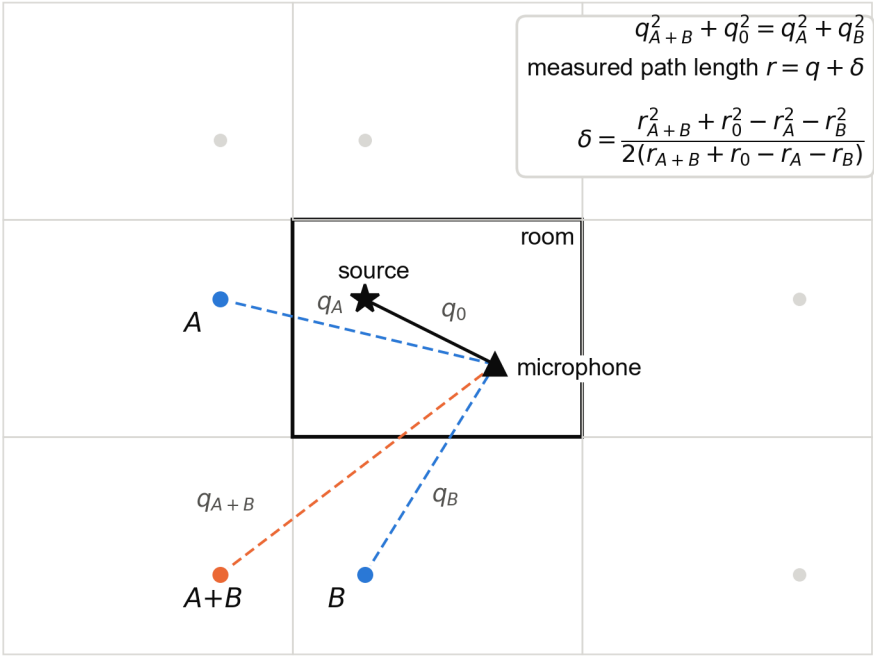



**Fig. 1.** The planar image-source lattice and the identity between images on different axes. The offsets of the side-wall image $A$, the end-wall image $B$ and their combination $A{+}B$ add axis by axis, so $q^2_{A+B} + q_0^2 = q_A^2 + q_B^2$; all four measured path lengths carry the same latency $\delta$.

### 3.3 Bearing and per-axis offsets

Let $c_i = s_i - m_i$ be the offset of the source relative to the microphone along axis $i$, so that $\sum_i c_i^2 = q_0^2$. A single-axis image, reflected only between the two walls of axis $i$, keeps the direct sound's offsets $c_j$ on the other axes, so its length is

$$\ell = \sqrt{\Delta^2 + q_0^2 - c_i^2}, \qquad \Delta \in \{U_i^-,\ U_i^+,\ 2L_i - c_i,\ 2L_i + c_i,\ 2L_i + U_i^-,\ 2L_i + U_i^+,\ \dots\},$$

where $U_i^- = 2m_i + c_i$ and $U_i^+ = 2L_i - 2m_i - c_i$ correspond to the two first-order walls. The two first-order walls give two equations for three unknowns, $m_i$, $L_i$ and $c_i$, and cannot determine $c_i$; an image reflected twice between the same pair of walls lies at $2L_i \mp c_i$ and supplies the third equation, consistent with the opposite-wall constraint of Moore et al. [9]. The offsets of the axes are then tied together by $\sum_i c_i^2 = q_0^2$: when no second-order reflection is picked on one axis, the $c_i^2$ of that axis is supplied by this

closure condition, but for one axis only. The source bearing is the direction of $\mathbf{c}/q_0$. First-order arrivals by themselves carry no information about the bearing: a corner reflection is fully determined by the two walls and the direct sound (the identity of the previous subsection), and any feasible bearing reproduces all the first-order arrivals; information about the bearing comes only from images reflected two or more times on some axis.

### 3.4 Generating candidate rooms

For each latency hypothesis, two generators propose candidate rooms; the full steps and constants are given as Algorithm 1 in Appendix A. `image_groups` joins echoes by edges according to the identity: echoes $a$ and $b$ are joined if an echo lies near $\sqrt{q_a^2 + q_b^2 - q_0^2}$. A single-axis image on one axis is joined to the single-axis images on every other axis, so starting from one early wall per axis (a joined pair in a planar room, three mutually joined echoes in three dimensions), the echoes joined to the walls of all the other axes form the group of that axis. Each group is fitted on a grid of $c_i$ with 5 mm spacing: the shortest member is taken as one first-order wall and paired with one of the next three members, which is read either as the other first-order wall or as the image at $2L_i \mp c_i$; when the group of an axis does not reveal $c_i$, it is given by the closure condition. `axis_scan` is used only in planar rooms: on the grid of $c_i$, every pair of echoes is scored as evidence for one axis (the score does not depend on which axis, so one table serves both axes), the best few pairs of each axis are combined on a 1° bearing grid, and the latency is also fine-tuned within ±4 cm in steps of 2 cm. By default both generators are used in planar rooms and only `image_groups` in three dimensions. Like Jager et al. [37], `image_groups` handles echo labelling on a graph, but according to that paper's abstract its graph is solved for multi-microphone labelling by a maximum independent set; here the edges come from the identity within a single response, and the echoes kept are those joined to the seeds of the other axes, with no maximum independent set.

### 3.5 Poisson point-process scoring

The picked echoes (including the direct sound) are treated as a realization of an inhomogeneous Poisson point process over the picking range [47]. The picking range $T$, in path length, runs from 3 samples before the first peak to 3 samples after the last; predicted arrivals outside it do not enter the likelihood. The intensity and the log-likelihood are

$$\lambda(r) = \frac{n_c}{T} + \sum_{\mathbf{k}} p_{\mathbf{k}}\, \mathcal{N}(r;\, r_{\mathbf{k}},\, \sigma^2), \qquad \log \mathcal{L}(\theta) = \sum_{j=1}^{N} \log \lambda(r_j) - \Big(n_c + \sum_{\mathbf{k}} p_{\mathbf{k}}\Big).$$

$n_c$ is the expected number of spurious peaks within the range (3 by default), $\sigma$ is the kernel width, and $p_{\mathbf{k}}$ is the expected number of peaks contributed by image $\mathbf{k}$. With $x$ the level in decibels of the predicted strength $\bar{\beta}^{n_{\mathbf{k}}} q_0/q_{\mathbf{k}}$ relative to the picking threshold (2% of the direct sound), $p_{\mathbf{k}} = 0.9/(1 + e^{-x/2})$, a logistic step with a ceiling of 0.9 and a width of about 4 dB, where $\bar{\beta} = 0.8$ is an assumed mean reflection coefficient and $n_{\mathbf{k}}$ is the number of reflections; arrivals less than one sample apart count once, with the largest of their $p_{\mathbf{k}}$. Spurious peaks are absorbed by the clutter term and missed arrivals by $p_{\mathbf{k}}$,

so the scoring needs no labels. This is a Poisson approximation: one predicted arrival can absorb several peaks, and the model does not require each image to produce at most one peak. Missing a strong image costs only $p_{\mathbf{k}}$, which is weaker than a model that describes each image by a Bernoulli detection, and this may be one reason why removing the expected-count term had no effect in Section 5.4. The minimum spacing of 2 samples in peak picking is a hard-core repulsion, and the merging of nearby arrivals is only an approximation; neither has a counterpart in the Poisson model.

The rationale for the expected-count term is this: a room with halved sides contains every true image, so a score that rewards only "explained echoes" would prefer it; here such a room must pay $p_{\mathbf{k}}$ for every image that is predicted to be strong but was not picked. In the 200 mismatch-free evaluation scenarios, however, removing this term did not change the outcome of any scenario (Section 5.4), so this rationale found no support in the evaluation of this paper.

### 3.6 Fitting

The best few distinct candidates (8 in planar rooms, 12 in three dimensions) are refined by EM [48]: each echo is shared among nearby predicted arrivals according to its responsibilities, and all parameters are then updated at once by weighted least squares. The kernel width $\sigma$ takes 6, 3, 1.5 and 0.8 cm in turn (never less than half a sample), so that the fit sees a smoothed likelihood first and a sharp one later, as in graduated non-convexity [49]. A second chain runs only the last two widths from the same starting point, so that a good starting point is not smoothed away; both chains keep the best point they visit. Next, ordinary least squares on only the echoes that clearly correspond to a single arrival removes the millimetre-level bias that soft responsibilities leave when images crowd together. Finally, with the other parameters fixed, each side length is rescanned on a 1 cm grid, and each microphone coordinate is rescanned with the source offset held fixed, which frees fits that found the right walls but pushed the microphone onto a wall.

### 3.7 Standard errors and the reliability flag

Every fit reports standard errors from the Laplace approximation [46], with covariance $\hat{\sigma}^2 (J^\top J)^{-1}$, where $J$ is the Jacobian of the matched arrivals with respect to the parameters and $\hat{\sigma}^2$ is their residual variance. They are conditional on the chosen correspondence between echoes and arrivals and do not include the uncertainty of the correspondence itself, so they describe only the local precision when the lattice is right; the output includes standard errors for the side lengths and the latency, but not for the bearing. Reliability is judged by two fractions: `explained_fraction` is the fraction of picked echoes within max(2 samples, 2 cm) of some predicted arrival, and `missed_fraction` is the fraction of arrivals predicted to be strong ($p_{\mathbf{k}}$ at least half of its maximum) that have no echo within the same tolerance. A fit is flagged reliable when both pass their thresholds: at least 0.9 for the former and at most 0.2 for the latter in planar rooms, and 0.9 and 0.1 in three dimensions. Both fractions use fixed thresholds; they do not use the likelihood of Section 3.5 and have no random baseline, and when peaks are dense even a wrong lattice can explain most of them (Section 6.2). The thresholds and the detection model were chosen on simulated development rooms and frozen before the evaluation (Section 4.3).

### 3.8 Identifiability and equivalent layouts

A single omnidirectional response cannot distinguish certain layouts. Along an axis of length $L_i$, the set of image offsets is invariant under four transformations: identity, reflection ($m_i \to L_i - m_i$, $s_i \to L_i - s_i$), swapping the source and microphone coordinates on that axis, and reflection followed by swapping; these transformations also preserve the number of reflections of every image. Applying them independently on each axis, together with relabelling the axes, gives $4^2 \times 2! = 32$ layouts that produce exactly the same response in a planar room (when all walls have the same reflection coefficient), and by the same argument $4^3 \times 3! = 384$ in three dimensions. Yu and Kleijn [10] list a 96-fold degeneracy in their three-dimensional setting; the two counts include different transformations, and this paper has not matched them item by item. This paper therefore reports only a canonical form: the side lengths are sorted from longest to shortest, the position is taken from the layout in which the source lies beyond the microphone on every axis, and the bearing is expressed in the sorted axes (in planar rooms as the angle from the longest side, in three dimensions as an azimuth and an elevation); the offset along each axis is determined only in magnitude, so the bearing lies within 0–90°. In three dimensions a single omnidirectional microphone cannot tell which side is vertical, so the "height" can only be taken by convention as the shortest side. These symmetries are of the same kind as the degeneracies noted by Moore et al. [9], so success is judged by the error of the sorted side lengths and does not depend on the choice of layout. The method has several components that can be removed separately; the experiments of the next section test them one by one with a paired design and ablations.

## 4. Experimental design

### 4.1 Simulation and measurement chain

The simulation generates responses with the image-source method [2]: the amplitude of each image is the reflection coefficient raised to the number of reflections, divided by the path length, and arrivals at non-integer samples are represented by Hann-windowed sinc pulses [50]. The lattice of a planar scenario contains only the two horizontal axes, which corresponds to a room whose floor and ceiling absorb completely and whose source and microphone are at the same height; this is an idealized situation (Section 6.1). Planar scenarios use a sampling rate of 16 kHz, simulated image orders up to 4 on each axis and a reflection coefficient of 0.85 for every wall; the excitation is 2 s of white noise, white noise at 30 dB is added, and a 1600-tap response is identified with a normalized least-mean-squares (NLMS) adaptive filter. Three-dimensional scenarios use 48 kHz, order 5 and a 1 s excitation, and identify 4800 taps by least squares. Signal-to-noise ratios in this paper always refer to the recording: the ratio between the power of the microphone signal during the excitation and that of the added white noise. Identification averages over the whole excitation, so the peak-to-noise ratio of the identified response itself (the ratio between the peak and the root mean square of the identification error) is much higher: in an exploratory calculation after review, the medians for E3 at 0, 10, 20, 30 and 40 dB were about 30, 40, 50, 60 and 70 dB (100 responses each). The identified responses are peak-picked by the rule of Section 3.1.

The scenario population is as follows: planar sides of $U(3, 10) \times U(3, 8)$ m, with a height of $U(2.4, 4)$

m added in three dimensions; the microphone and the source are sampled uniformly at least 0.3 m from the walls, with a distance between them of 0.6–2.5 m; the latency is sampled from $U(0, 20)$ ms. The evaluation scenarios are separate from those used during development: a new master seed, 20261024, derives the scenario and noise seeds of every scenario.

Both the inversion model and the simulator are based on the image-source method, and generating and inverting data with the same model may overstate a method's performance [51]; the method is therefore also tested under model mismatch (E5), distribution shift (E9) and real measurements (E10). E5 is only a sensitivity analysis under mild mismatch: it has no scattering, no frequency-dependent absorption, and no transducer responses or directivity, and its strongest clutter condition has only 5 spurious pulses, so it cannot predict performance on real measurements.

## 4.2 Compared methods

The same identified response is passed to each method of Table 2, which forms a paired comparison. The first-order closed-form solver (M3–M5) is a baseline the author implemented in the same code base, not a reproduction of a published method: it takes the earliest 9 arrivals as the direct sound, 4 wall reflections and 4 corner reflections, solves for the room in closed form for a given bearing, and checks the echo assignment with the corner reflections; it applies only to planar rooms. M8 is an experiment added after review. Following Section 3.4 of Moore et al. [9], it takes the labels of M7, keeps only images up to second order, and minimizes the squared error of all time differences of arrival: the position of the source relative to the microphone is searched on a grid with a 3 m radius and 10 cm spacing, at each grid point the distances to the four walls are found by trust-region reflective least squares (starting from Eqs. (16)–(17) of that paper), and the best grid point is then refined over all six parameters with the same cost; it is used only in E1.

**Table 2.** Compared methods

| Label | Method | Purpose |
|---|---|---|
| M1 | Lattice inversion with default settings, estimating both latency and bearing | Proposed method |
| M2 | Lattice inversion given the true latency | Cost of estimating the latency |
| M3 | First-order closed-form solver, bearing assumed to be 45°, unsynchronized input | The author's first-order baseline on unsynchronized input |
| M4 | First-order closed-form solver, input first aligned with the true latency, bearing 45° | First-order closed form under the synchronization it needs |
| M5 | First-order closed-form solver, aligned and given the true bearing | Upper bound of the first-order closed form |
| M6 | Same scoring and EM fitting as M1, starting points sampled at random in parameter space, same wall-clock time as M1 | Whether the latency hypotheses and candidate generators are needed |

| Label | Method | Purpose |
|---|---|---|
| M7 | Picked echoes assigned to the nearest true image arrival within max(2 samples, 2 cm) (one per image, the rest treated as spurious), then the same least squares as in the refinement of M1, keeping the smallest residual from 5 starting points at the truth plus perturbations (±10 cm in positions and side lengths, ±3 cm in latency) | Upper bound with known labels and a starting point near the truth |
| M8 | Time-difference least-squares method of Moore et al. [9], labels taken from M7, images up to second order only (added after review, E1 only) | Performance of a published method with known labels |

### 4.3 Experiments and analysis

The experiments, the definition of success and the analyses were all written down before the evaluation and are reported following the reporting principles for simulation studies [52]; Table B1 in Appendix B gives an overview of the experiments. E1 is the planar main experiment, with 500 scenarios. E1b uses the first 100 scenarios of E1 and simulates only first-order reflections with synchronized recording, to test the first-order closed-form solver within its own scope; E1t times the first 20 scenarios in a single process. E2 is the three-dimensional main experiment, with 200 scenarios. E3–E5 and E9 each sweep one condition over 100 planar scenarios: the scenarios are derived from the master seed by the name of each experiment and differ from those of E1, and all levels of one experiment share the same 100 scenarios. The model mismatches of E5 are wall-by-wall reflection coefficients $U(0.6, 0.95)$ (R1), simulated order 8 (R2), simulation with the speed of sound at 15 °C and 30 °C and inversion with 343 m/s (R3), 5 spurious pulses with amplitudes of 3–10% of the direct sound (R4), and sampling-rate skews of 20 and 100 ppm between playback and recording (R5). The distribution shifts of E9 are small rooms of $U(2, 4)^2$ m, large rooms of $U(10, 20) \times U(8, 15)$ m, source distances of 2.5–5 m and microphones 0.05–0.3 m from a wall. E6 runs six ablations on the first 200 scenarios of E1 (Table 4); its M1 row is the result of E1 on these 200 scenarios and was not rerun. E7 and E8 pool the results above to analyse the reliability flag and the standard errors. The numbers of scenarios were set in advance from the expected precision: at a success rate of about 0.95, the 500 scenarios of E1 give a Wilson 95% interval with a half-width of about 2 percentage points (a Monte Carlo standard error of about 1 percentage point), and the 200 three-dimensional scenarios give a half-width of about 7 percentage points at a success rate of about 0.55. The sweep conditions have 100 scenarios each, with half-widths of about 4–10 percentage points, so two conditions a few percentage points apart (for example 98% and 94%) cannot be told apart. The development scenarios, the choice of thresholds and the results on the development set are given in Appendix C.

Success is defined as a summed absolute error of the sorted side lengths, $e = \sum_i |\hat{L}_{(i)} - L_{(i)}|$, below 5 cm, which in planar rooms corresponds to about one sample of path per side. A run that finds no solution always counts as a failure. The primary results, specified in advance, are the success rate of M1 in E1 (P1) and the precision and failure detection rate of the reliability flag in E1 (P2): precision is the fraction of

fits flagged reliable that succeed, and the failure detection rate is the fraction of failed fits that are flagged unreliable. Proportions are reported with Wilson 95% confidence intervals (CI); paired comparisons on the same scenarios use the two-sided exact McNemar test, with the six ablations corrected by the Holm method; medians are reported with percentile intervals from 10,000 bootstrap resamples. The sweep experiments report only estimates and intervals, without further tests. The detection model, the reliability thresholds and the other settings were frozen before the evaluation; every result records the git commit and source hash of the code, and the hashes of the configuration files are recorded in the freeze file `freeze.json` [53], whose release is described in the section on data and code availability. Experiments and analyses added after review are numbered PR1–PR13 (post-review, listed in Table B1 in Appendix B); none of them changed any frozen setting, and apart from PR13, which compared peak-picker settings and clutter counts on microphones that E10 did not use, nothing was tuned on dEchorate.

### 4.4 Real measurements

E10 uses the dEchorate dataset [12]: a cuboid room of about 6 × 6 × 2.4 m in which every surface except the floor, which is always absorbing, can be switched between absorbing and reflecting. It is measured at 48 kHz, and the dataset paper gives a speed of sound of 346.98 m/s for the 24 °C and 80% relative humidity on site [12]. The pre-specified subset is 10 empty-room wall configurations × 9 sources × 6 microphones (the middle one of each array), 540 responses in all, plus 54 responses with furniture. The sources are 4 directional loudspeakers facing the centre of the room and 2 facing walls [12], and 3 omnidirectional loudspeakers whose positions in the data files are not calibrated. The ground truth is the calibrated geometry of the dataset authors (room 5.705 × 5.965 × 2.355 m); the primary analysis takes the speed of sound from the formula of the calibration script at 24 °C, 345.844 m/s, and the secondary analysis takes 346.98 m/s. The measurements themselves are synchronized, so each response is given an artificial latency drawn from $U(0, 20)$ ms (with a seed fixed in advance), and the onset of the responses is first checked against the calibrated geometry with the 4 loudspeakers facing the room. The inversion uses the same three-dimensional settings as E2, frozen before any response was read, and no parameter was adjusted on dEchorate. The frozen detection model predicts whether an image is picked with a $\bar{\beta}$ shared by all surfaces, which does not match the dEchorate configurations with an absorbing floor and some absorbing walls; the effect of this is reported configuration by configuration in Section 5.6. The dataset annotates only the direct sound and the first-order echoes [12], so the labels of M7 are instead given by the image arrivals computed from the calibrated geometry (orders up to 5 on each axis).

There are two deviations from the original plan. First, the pre-specified v2.0 response file was already corrupted at the source and could not be read, so the author switched to the SOFA (Spatially Oriented Format for Acoustics) version regenerated by the dataset authors, whose responses are not bit-identical to v2.0. Second, the wall configurations could not be confirmed from the attributes of the SOFA files, so before inversion the configurations given in the file names were checked against the response strength near first-order images, and the check failed (Section 5.6). The change of file and the secondary speed of sound were decided before any response was decoded, and the handling after the failed check was decided before any inversion was run: E10 was run with the frozen settings, and the groupings that

depend on the south-wall label are marked unverified. The next section reports the results in the order of the sub-questions.

# 5. Results

## 5.1 Planar main experiment

Table 3 lists the results of the methods in E1. M1 succeeded in 489 of the 500 planar rooms (97.8%, 95% CI 96.1–98.8%), with a median summed side error of 1.26 mm (1.18–1.33 mm); over all 500 fits the median latency error was 0.0039 ms and the median bearing error 0.038°. The error distribution is bimodal: 97.8% of the fits had errors below 1 cm, while all 11 failures had errors above 20 cm, 5 of them above 1 m, so the conclusions do not depend on the choice of the 5 cm threshold. The 11 failed scenarios are numbers 41, 95, 96, 121, 152, 163, 181, 196, 198, 223 and 478, 9 of them within the first 200. After review, the responses were rebuilt with the same master seed and M1 was rerun; the estimates and outcomes of all 500 scenarios matched bit for bit, so the concentration in the early scenarios is not a difference between runs. The seed of each scenario is derived independently, and this paper has not found the cause of the concentration. M2, given the latency, succeeded in all 500; every discordant scenario was one where M2 succeeded and M1 failed (0 versus 11, exact McNemar p = 0.00098), so estimating the latency costs 2.2 percentage points. M7, with known labels, also succeeded in all, with a median error of 1.24 mm; in simulation, the median number of the 40 picked peaks that could be assigned to image arrivals was 40, that is, all of them; on real measurements the picture is entirely different (Section 5.6).

**Table 3.** Results of the methods in E1 (500 planar rooms). Success means a summed error of the sorted side lengths below 5 cm; intervals are Wilson 95% confidence intervals; M6 uses only the first 200 scenarios; the flag is reported only for M1, as specified in advance; M8 was added after review.

| Method | Successes | Success rate (95% CI) | Median error | Reliability flag |
|---|---|---|---|---|
| M1 lattice inversion | 489/500 | 97.8% (96.1–98.8) | 1.26 mm | Precision 489/489; failure detection 11/11 |
| M2 latency known | 500/500 | 100% (99.2–100) | 1.23 mm | — |
| M3 first-order, 45°, unsynchronized | 0/500 | 0% (0–0.8) | 11.8 m | — |
| M4 first-order, aligned, 45° | 2/500 | 0.4% (0.1–1.4) | 9.88 m | — |
| M5 first-order, aligned, true bearing | 8/500 | 1.6% (0.8–3.1) | 8.85 m | — |
| M6 random starts | 2/200 | 1.0% (0.3–3.6) | 2.08 m | — |
| M7 known labels (upper bound) | 500/500 | 100% (99.2–100) | 1.24 mm | — |
| M8 Moore et al. [9], labels from M7 (added after review) | 500/500 | 100% (99.2–100) | 3.5 mm | — |

The author's first-order closed-form solver failed on almost all of the same responses: the success rates of M3, M4 and M5 were 0%, 0.4% (2/500) and 1.6% (8/500), with median errors of 11.8 m, 9.88 m and 8.85 m. The cause lies in the picked echoes: the simulation contains higher-order reflections, and of the 9 arrivals picked by M4 and M5 only 5.6 on average were true first-order arrivals, with all 9 correct in only 1 scenario. Within the first-order closed-form solver's own scope (E1b: only first-order reflections simulated, synchronized recording), the success rate of M5 recovered to 63% (63/100, 53.2–71.8%), and that of M4 was 17%; M1 could not run, because there were only 9 echoes, fewer than the 10 it needs, and this was recorded as a failure.

M8, added after review, applies the method of Moore et al. [9] to the same 500 responses, with labels taken from M7 and only images up to second order (a median of 13 per response, at least 9). M8 succeeded in all of them (500/500, 99.2–100%), with a median summed side error of 3.5 mm; it agreed with M7 and M2 in every scenario, and the 11 scenarios where it disagreed with M1 were all M8 successes (11 versus 0, exact McNemar $p = 0.00098$). With known labels, the existing time-difference least-squares method already reaches the upper bound; what lattice inversion adds is that it needs no labels, at a cost of these 2.2 percentage points.

M6, with random starting points under the same scoring, fitting and wall-clock time (a median of 782.5 starting points), succeeded in only 2/200 (1.0%, 0.3–3.6%), with a median error of 2.08 m. Scoring and fitting alone are not enough to find the right lattice; the closed-form latency hypotheses and the candidate generators provide the starting points that are needed. For timing (E1t, single process, 20 scenarios), the median wall-clock time per response was 5.19 s for M1, 0.40 s for M2 and 0.036 s for M3.

### 5.2 Three dimensions

Three-dimensional rooms (E2) are much harder. M1 succeeded in 112 of 200 rooms (56%, 49.1–62.7%) and M2, given the latency, in 162 (81%, 75.0–85.8%), with 12 versus 62 discordant scenarios ($p = 2.9 \times 10^{-9}$); M7, with known labels, succeeded in all, and M6, with random starting points, failed in all 100 rooms. The errors of M1 were again bimodal: 53.5% of the fits had errors below 1 cm and 40% above 1 m (including 1 run that found no solution); over all 200 fits the median latency error was 0.0041 ms and the median bearing error 0.13°. Again the median number of the 120 picked peaks that could be assigned to image arrivals was all 120. In three dimensions estimating the latency costs 25 percentage points, far more than in planar rooms. M7 succeeded in all, but it was given both the labels and a starting point near the truth (Section 4.2), so it shows only that fitting precision is not the bottleneck under these two conditions; the difficulty lies in finding the right lattice when neither the labels nor the starting point is known, and the present design cannot separate these two factors. For reference, Yu and Kleijn [10] report a failure rate of 0.7% in three-dimensional simulations with a known emission time and no scattering, and 18.9% at a scattering coefficient of 0.2; their simulator and the echoes they use differ from those of this paper, so this comparison is for reference only.

### 5.3 Sensitivity

Fig. 2 summarizes the success rates of the planar sweep experiments. The recording signal-to-noise ratio (E3) has little effect at 20 dB and above: the success rate of M1 was 96%, 95% and 94% at 40, 30 and 20 dB, fell to 12% at 10 dB, and was 0% at 0 dB; the corresponding peak-to-noise ratios after identification were about 70, 60, 50, 40 and 30 dB (Section 4.1). The first-order closed-form solver M5 reached only 0–2% at every signal-to-noise ratio. With 100 scenarios per condition, the interval half-widths are about 4–10 percentage points, so levels a few percentage points apart (for example 96% and 94%) cannot be told apart. The latency (E4) had almost no effect between 0 and 40 ms (98%, 96%, 97% and 96% in turn), and success fell to 83% at 150 ms; M2 was at 100% at every latency. The 150 ms condition also switched identification to least squares with 4000 taps, so the present design cannot attribute this drop to the size of the latency, and lattice inversion is mathematically equivariant to the absolute size of the latency.

Under model mismatch (E5), wall-by-wall reflection coefficients (R1) gave 98%, simulated order 8 (R2) 95%, 5 spurious pulses (R4) 87%, a sampling-rate skew of 20 ppm (R5) 94%, and 100 ppm 50%. A sampling-rate skew stretches the response gradually over the 2 s of identification: 100 ppm accumulates 0.2 ms in 2 s (about 3.2 samples at 16 kHz), whereas 20 ppm gives only about 0.6 samples, so the latency is no longer a single fixed value; this is a possible mechanism, which this paper has not verified separately. An error in the speed of sound (R3) gave a success rate of 0% at 5 cm, but this is the proportional scaling expected in advance: the simulation takes the speed of sound as $c(T) = 331.3\sqrt{1 + T/273.15}$ m/s, so inverting with 343 m/s should scale the dimensions by $343/c(T)$, that is, by +0.80% and −1.72% at 15 °C and 30 °C, which matches the observed median relative dimension errors of 0.80% and 1.72% (in absolute value). Under distribution shift (E9), success was 87% in small rooms, 98% in large rooms, 95% with a distant source and 79% with the microphone near a wall. All these mismatches are mild (Section 4.1).

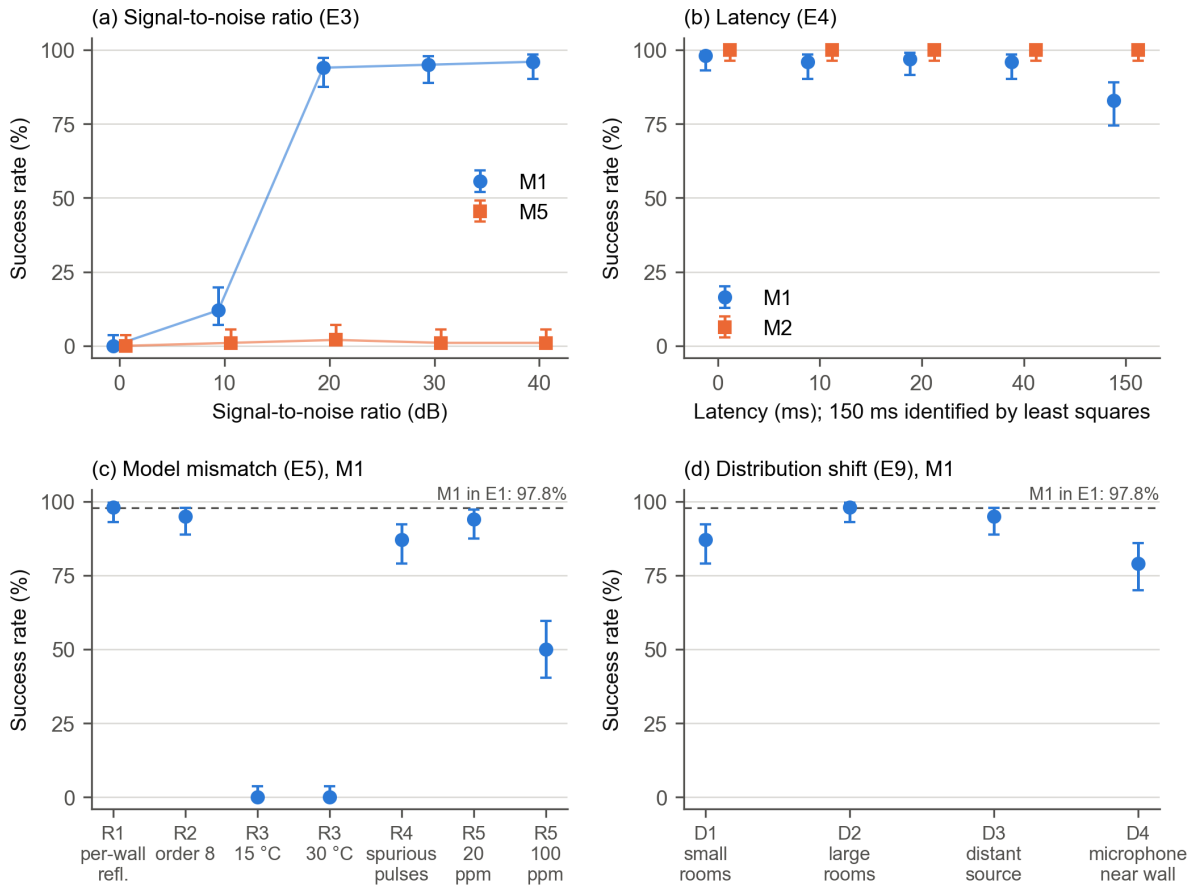



**Fig. 2.** Success rate of M1 in the planar sensitivity experiments, with Wilson 95% confidence intervals: (a) signal-to-noise ratio (E3, with M5 also shown), (b) latency (E4, with M2 also shown), (c) model mismatch (E5), (d) distribution shift (E9). 100 scenarios per condition.

### 5.4 Ablations

Table 4 lists the ablation results of E6. The M1 row is the result of the first 200 scenarios of E1 (191 successes, 95.5%) and was not rerun; 9 of the 11 failures of E1 fall within these 200 scenarios, so its success rate is lower than that of E1 as a whole. Removing the expected-count term (A1), using a single annealing chain (A2), skipping the matched least squares (A3) or skipping the per-axis rescan (A4) all kept the success rate at 95.5%, each with 0 scenarios discordant with M1; these variants did change some estimates (A3, for example, changed the estimates of 187 scenarios), but not the outcomes. A5, which uses only `image_groups`, matched M1 (1 versus 1 discordant) and was much faster: with 14 processes running in parallel, the median wall-clock time per response was about 2.4 s for A5 and 42–44 s for the other five variants (these are not single-process timings). A6, which uses only `axis_scan`, lost 8 scenarios (91.5%, 8 versus 0, Holm-corrected p = 0.047) and was the only ablation with a measurable effect.

**Table 4.** E6 ablations (first 200 scenarios of E1). The discordant column gives the numbers of scenarios where only M1 succeeded / only the variant succeeded; p values are from the exact McNemar test with Holm correction (six ablations).

| Variant | Successes | Success rate (95% CI) | Discordant | Holm-corrected p |
|---|---|---|---|---|
| M1 (full) | 191/200 | 95.5% (91.7–97.6) | — | — |
| A1 expected-count term removed | 191/200 | 95.5% (91.7–97.6) | 0/0 | 1.00 |
| A2 single annealing chain | 191/200 | 95.5% (91.7–97.6) | 0/0 | 1.00 |
| A3 no matched least squares | 191/200 | 95.5% (91.7–97.6) | 0/0 | 1.00 |
| A4 no per-axis rescan | 191/200 | 95.5% (91.7–97.6) | 0/0 | 1.00 |
| A5 image_groups only | 191/200 | 95.5% (91.7–97.6) | 1/1 | 1.00 |
| A6 axis_scan only | 183/200 | 91.5% (86.8–94.6) | 8/0 | 0.047 |

### 5.5 Reliability flag and standard errors

In E1 the reliability flag separated successes from failures completely: all 489 fits flagged reliable succeeded (precision 100%, 99.2–100%), and all 11 failures were flagged unreliable (failure detection rate 100%, 74.1–100%). With only 11 failures, the lower bound of the interval for the failure detection rate is only 74.1%; three dimensions, with more failures, give a more precise reference. In three dimensions (E2) the precision was 94.9% (112/118, 89.3–97.6%) and the failure detection rate 93.2% (82/88, 85.9–96.8%): again no successful fit was flagged unreliable, but 6 wrong fits were flagged reliable. The third cell, not specified in advance, is the fraction of successful fits flagged reliable; it was 100% in both E1 and E2 (489/489, 112/112). The two P2 measures do not penalize over-rejection, and this cell covers that point.

Pooling the 1,700 fits of E1, E3 and E5 (E7), the precision of the flag fell to 0.79 (1,122/1,424) and the failure detection rate was only 0.38 (188/490). The same scenarios recur under different conditions, so the intervals were computed by a bootstrap clustered by scenario (700 clusters): 0.77–0.80 and 0.35–0.42. The pooled values depend on the weight of each condition, and the per-condition $2 \times 2$ tables are provided with the result tables. The fraction of successful fits flagged reliable (not specified in advance) was 92.7% pooled (1,122/1,210, 91.2–94.2%). The degradation is concentrated in two conditions. At 0 dB all 100 fits failed, and 96 of them were flagged reliable; the medians of `explained_fraction` and `missed_fraction` for these 100 fits were 0.975 and 0.062, both within the planar thresholds (descriptive medians taken from the result files, as below). At 10 dB the two medians were 0.725 and 0.309; the flag marked 86 of the 88 failures, but all 12 successful fits were also flagged unreliable (0/12), so the failure of the flag appears when the signal-to-noise ratio falls to 0 dB. With an error in the speed of sound (R3), 98 and 97 fits were flagged reliable, although all of them count as failures at 5 cm. Wall-by-wall reflection coefficients (R1), spurious pulses (R4) and a sampling-rate skew of 100 ppm caused some successful fits to be flagged unreliable (85/98, 57/87 and 21/50 of the successes were flagged reliable). In all four groups of the distribution shift (E9), every fit flagged reliable succeeded.

The Laplace standard errors are on the conservative side in planar rooms. Conditioned on being flagged reliable (in E1, the 489 successful fits), the fractions of standardized errors $z$ within $\pm 1.96$ were 98.8% (97.4–99.4%) and 97.8% (96.0–98.7%) for the two sides and 99.2% (97.9–99.7%) for the latency, above

the nominal 95% (E8; the standard deviations of $z$ for the two sides were 0.76 and 0.81). In three dimensions they tend to be too small: conditioned on being flagged reliable (118 fits, including 6 failures), the three sides gave 84.7% (77.2–90.1%), 83.1% (75.3–88.8%) and 82.2% (74.3–88.1%), and the latency 88.1% (81.1–92.8%); counting only the 112 successful fits, the three sides gave 86.6%, 84.8% and 85.7%. The coverage conditioned on being flagged reliable is an exploratory calculation after review, and it is exactly the set of fits that users see.

### 5.6 Real measurements

Table 5 lists the results of E10. On the 540 empty-room responses, M1 and M2, which is given the latency, failed every time (0/540, 95% CI 0–0.7%), with median summed side errors of 11.4 m and 10.1 m; M1 found no solution 179 times and M2 in only 1 case, and the other fits converged to wrong rooms. The 54 responses with furniture were likewise 0/54, and the results were the same with a speed of sound of 346.98 m/s. The reliability flag passed none of the M1 or M2 fits, but it also rejected every M7 fit, including the 162 that succeeded: on measured responses the flag could not tell correct fits from wrong ones. The rejections come from `missed_fraction`: the 361 M1 fits that found a solution explained a median of 95.8% of the peaks, but 28.7% of the predicted strong images had no corresponding peak, above the three-dimensional threshold of 0.1. The intervals in the table describe only the sampling variation within this one room; with a bootstrap clustered by source position (9 clusters), the interval for M7 is 21.9–38.0%, and the individual sources range from 11.7% to 51.7%. The frozen onset check used only the 4 loudspeakers facing the room: without the artificial latency, the median difference between the response maximum and the calibrated direct sound was 0.0038 ms at the primary speed of sound, less than one sample, and was not compensated further; at the secondary speed of sound it was 0.0251 ms, more than one sample, and was absorbed into the true latency under the frozen rule. After review, the peak that the picker took as the direct sound was checked separately for each source type: the median difference from the calibrated direct sound was 0.016 ms for the loudspeakers facing the room (161 of 240 within one sample), 0.22 ms for those facing a wall (at most 7.8 ms) and 1.49 ms for the omnidirectional loudspeakers, and in neither of the latter two types was any response within one sample. For the 300 responses from the loudspeakers facing a wall and the omnidirectional loudspeakers, the premise of Section 3.1 that the first peak is the direct sound does not hold, or the ground truth itself is wrong.

The dataset authors report a mean inconsistency of 0.4 cm and a maximum of 1.86 cm between the calibrated positions in geometric space and in signal space, but the share of first-order echo times that agree with the geometry is 98.1% within 0.5 ms, only 53.1% within 0.1 ms and 30.2% within 0.05 ms, and the uncertainty of the room side lengths is not reported [12]. M7, labelled from the calibrated geometry, came within 5 cm in only 162/540 (30%, 26.3–34.0%), with a median error of 8.2 cm; relaxed to 10 cm and 15 cm, the rates were 60% and 75%. With 346.98 m/s it was 33% (176/540, 28.8–36.7%), with a median error of 7.7 cm. By source type, the directional loudspeakers facing the room gave 40% (97/240), those facing a wall 12.5% (15/120), and the omnidirectional loudspeakers with uncalibrated positions 28% (50/180); counting only the 6 calibrated directional loudspeakers, the rate was 31% (112/360). The 54 responses with furniture gave 33% (18/54), and 20% (11/54) at the secondary speed of sound; across

wall configurations the rates ranged from 22% to 37% (with the configurations that depend on the south-wall label unverified). Of the 120 peaks of each response, M7 could assign a median of only 25 to image arrivals.

**Table 5.** Results of E10 (dEchorate). Primary speed of sound 345.844 m/s, secondary 346.98 m/s; 540 empty-room responses and 54 with furniture; intervals are Wilson 95% confidence intervals; the last column is the number of empty-room fits flagged unreliable.

| Method | Speed of sound (m/s) | Empty-room successes | Success rate (95% CI) | Median error | Successes with furniture | Flagged unreliable |
|---|---|---|---|---|---|---|
| M1 lattice inversion | 345.844 | 0/540 | 0% (0–0.7) | 11.4 m | 0/54 | 540/540 |
| M2 latency known | 345.844 | 0/540 | 0% (0–0.7) | 10.1 m | 0/54 | 540/540 |
| M7 known labels | 345.844 | 162/540 | 30% (26.3–34.0) | 8.2 cm | 18/54 | 540/540 |
| M1 lattice inversion | 346.98 | 0/540 | 0% (0–0.7) | 11.4 m | 0/54 | 540/540 |
| M2 latency known | 346.98 | 0/540 | 0% (0–0.7) | 10.0 m | 0/54 | 540/540 |
| M7 known labels | 346.98 | 176/540 | 33% (28.8–36.7) | 7.7 cm | 11/54 | 540/540 |

After review, the results were reported separately for each of the 10 wall configurations (Table D1 in Appendix D), and the configurations did not differ: configurations in which both walls of one or two axes reflect also failed every time, and the `missed_fraction` of the ground truth was about 0.3 in every configuration (the configurations that depend on the south-wall label remain unverified). Scored with the frozen detection model, the calibrated ground truth has a lower Poisson log-likelihood than every M1 fit (361) and every M2 fit (539) that found a solution, by a median of about 347; the ground truth would be flagged unreliable on every response. This comparison does not depend only on the accuracy of the ground truth: the M7 fits, labelled and refined near the ground truth, also have a lower likelihood than every M1 and M2 fit on the same response (by a median of about 335), and this holds even for the successful M7 fits alone (103 with an M1 fit to compare, by a median of about 306). These fits were reconstructed from stored values with four decimal places; the two fractions after reconstruction differ from the stored values by at most 0.015 (M1, M2) and 0.042 (M7).

An exploratory diagnosis that was not pre-specified takes every 11th response of the E10 subset, 54 measured responses (including 4 with furniture and 18 from omnidirectional loudspeakers), and compares them with the first 20 simulated three-dimensional responses (Fig. 3). What does not depend on errors in the ground truth is the count: the 120 peaks of a measured response span only 19.0 ms after the first peak (median), with a median of only 40 image arrivals in that range, whereas the 120 peaks of a simulated response span 36.8 ms, with a median of 144.5 image arrivals in range. The share of peaks that lie near an image arrival, on the other hand, cannot serve as evidence. The random baselines added after review show

that the per-response median shares for the measured responses at tolerances of 2 cm, 0.1 ms and 0.5 ms were 21.3%, 34.2% and 81.2% (1,496/6,438, 2,272/6,438 and 5,210/6,438 pooled over the 54 responses); with the peak positions redrawn uniformly within the picking range they were 22.0%, 34.6% and 83.0% (1,000 draws each; pooled expected counts 1,521.3, 2,318.5 and 5,297.5), and with a circular shift of all peaks 21.8%, 34.4% and 82.7% (1,509.2, 2,309.9 and 5,299.9), all the same as random. Excluding the 3 omnidirectional loudspeakers (36 responses) gives the same result (at 2 cm, a median of 20.9% versus 21.7%; pooled 1,014/4,278 versus an expected 1,049.8). For the simulated responses the per-response medians were 100% at all three tolerances (2,394/2,400, 2,400/2,400 and 2,400/2,400 pooled over the 20 responses), against uniform baselines of 34%, 48% and 86%. After review, an exploratory comparison that was not pre-specified also used the dataset's first-order echo annotations in Hierarchical Data Format 5 (HDF5) as a reference. The annotated arrival times all differ from the image arrivals of the calibrated geometry at 345.844 m/s by at most 2 ns, so they are not a reference independent of the geometry, and the counts based on either are identical. The annotations cover only the 6 directional loudspeakers; over the 2,160 first-order arrivals of these 360 empty-room responses, what was computed is whether a peak lies near each arrival (with peaks picked from the synchronized responses, without the artificial latency), a per-arrival share that differs from the per-peak shares above. Within 2 cm, 395/684 (57.7%) of the arrivals from reflecting walls and 836/1,476 (56.6%) of those from absorbing walls had a peak nearby, against expectations of 296.2 and 660.9 (43.3% and 44.8%) with peak positions redrawn uniformly; within 0.1 ms the shares were 528/684 (77.2%) and 1,121/1,476 (75.9%), against expectations of 412.4 and 922.0 (60.3% and 62.5%); within 0.5 ms they were 599/684 and 1,351/1,476, the same as the random 597.9 and 1,347.2. There are more peaks than chance near the first-order arrivals, but the shares for reflecting and absorbing walls are almost the same. The check of the wall labels used the ratio of the response strength near the first-order images to the direct sound: the minimum when the ceiling reflects, 0.662, is above the maximum when it absorbs, 0.246, and the same holds for the west wall (0.550 versus 0.318), but not for the south, east and north walls (0.190 versus 0.311, 0.374 versus 0.460, 0.288 versus 0.368). Compared pair by pair with the all-absorbing configuration, the south wall shows no change in any configuration in which it is labelled reflecting. The groupings that depend on the south-wall label are therefore unverified, but the wall labels do not enter the pooled results above. The high success rate in simulation and the complete failure on real measurements need to be explained together.

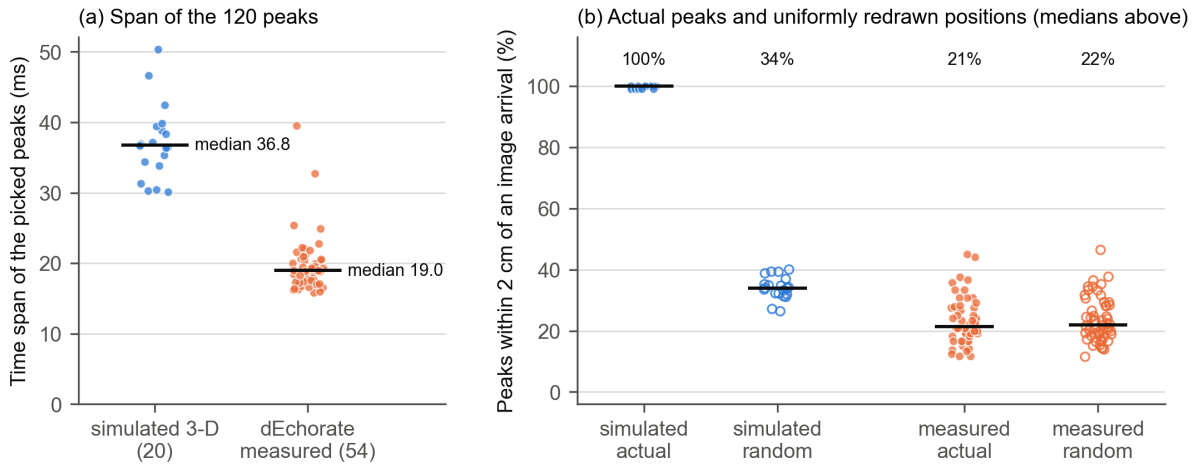


**Fig. 3.** Picked peaks and image arrivals: measured dEchorate responses (54) compared with simulated three-dimensional responses (20). (a) Time span covered by the picked peaks; (b) share of peaks within 2 cm of an image arrival, filled for the actual peaks and open for peak positions redrawn uniformly within the picking range (mean of 1,000 draws). Exploratory analysis, not pre-specified; the random baseline in (b) was added after review.

### 5.7 Diagnosis of the failure

To tell whether the failure on real measurements lies in the peak picker or in the data themselves, an exploratory diagnosis (PR13) was added after review. It checks three layers, the data, the peak picking and the inversion end to end, with the criteria of each written down before the responses it uses were read (Table E2); where the procedure departed from the plan is listed in Appendix E. The data come from microphones that E10 did not use: capsule 1 of each array for development and capsule 5 for confirmation, 360 responses each, with only the 6 directional loudspeakers whose positions are calibrated; the responses that E10 used were not read again.

For the 4 loudspeakers facing the room in the development set (240 responses), the first-order echoes were predicted from the calibrated geometry, and the maximum of the absolute response within ±0.25 ms of each prediction was taken (Table E1). When the ceiling reflects, the median offset is −0.4 cm and 67% lie within 2 cm, against 14% when the same search is run at random times; when the ceiling absorbs, only 17% (23% at random); the echo from the floor, which always absorbs, is systematically about 5.3 cm late; for the four side walls, reflecting or absorbing, the shares within 2 cm (14% and 20%) are no higher than at random times (17% and 21%). Taking the peak positions of all images up to second order with the same search and fitting the room with labels taken from the geometry, only 31% of the 360 development responses came within 5 cm in summed side error, and 80.0% within 10 cm; the confirmation set gave 28% and 79.7%. With a matched filter that uses the direct sound as its template, or with peaks of the response envelope (37 settings including the frozen picker), the share of picked peaks within 2 cm of an image arrival was 0.85–1.02 times the random baseline (the criterion is a ratio of 2). M1 inverted from 3 of these pickers brought none of the 240 facing-room responses within 5 cm and at most 1 within 10 cm, and flagged no fit as reliable (Table E2). The data layer passed only at the relaxed 10 cm, and only just; the peak-picking and end-to-end layers failed.

## 6. Discussion

### 6.1 Conditions under which the method works in simulation

In mismatch-free planar simulations, lattice inversion almost always recovers the side lengths, the latency and the bearing together, only 2.2 percentage points short of the two upper bounds (M2, M7): under this condition an unknown latency and unlabelled echoes cost little. With known labels, the method of Moore et al. [9] also succeeds every time (M8); what lattice inversion adds is only that it needs no labels. The planar results correspond to an ideal room whose floor and ceiling absorb completely and whose source and microphone are at the same height (Section 4.1); once the vertical axis is also put into the lattice,

the equally mismatch-free three-dimensional simulations reach only 56%. The planar capability rests on four conditions. First, the response must contain reflections of second or higher order: with only first-order reflections, lattice inversion does not even get the number of echoes it needs (E1b), and first-order arrivals carry no information about the bearing (Section 3.3). Second, the recording signal-to-noise ratio must be 20 dB or more (Section 5.3). Third, most of the picked peaks must be true arrivals: in simulation almost every picked peak corresponds to an image arrival, which is why M1 comes close to M7. Fourth, the sampling-rate skew between the two devices must be small: 94% at 20 ppm, falling to 50% at 100 ppm (Section 5.3). Two devices always have some sampling-rate skew, so in practice it must first be estimated and compensated, for example from the phase drift of the signal coherence function [54]. In three dimensions, Yu and Kleijn [10] likewise need no prior echo labels and report a failure rate of 0.7% with a known emission time (the simulators, echoes and definitions of failure differ, so this is for reference only); M2 of this paper, given the latency, still fails in 19%, whereas M7, given both the labels and a starting point near the truth, succeeds every time. The weakness of the three-dimensional search of this paper is thus in finding the lattice when neither the labels nor the starting point is known; the present design cannot separate these two factors (Section 5.2).

### 6.2 What the flag can and cannot do

The reliability flag checks only two things, how many of the picked peaks the lattice explains and how many of the expected images it misses, and both use fixed thresholds. An exploratory analysis after review redrew the peak positions uniformly within the picking range and computed the `explained_fraction` of the same M1 lattice: in E1 the actual median was 1.000 and the redrawn one 0.203; at 10 dB, 0.725 versus 0.365; at 0 dB, 0.975 versus 0.932; in E10, 0.958 versus 0.926. When peaks are dense, any lattice explains most of them, `explained_fraction` loses its power to discriminate, and only `missed_fraction` still acts in the flag. At 0 dB the wrong lattices rarely miss predicted images, so both fractions pass their thresholds (Section 5.5); on measured responses even the 162 successful M7 fits with known labels miss a median of about 32% of the strong images (28% over all 540 M7 fits), so all of them are rejected, and the flag cannot tell right from wrong on measured responses (Section 5.6). An error in the speed of sound, in turn, preserves the structure of the lattice: a wrong speed of sound scales the whole room proportionally, the scaled room fully explains the picked peaks, and a single response cannot tell such a scaling apart by itself. The flag has therefore been validated only within the simulated distribution; it can reject fits whose structure does not match, but it cannot guarantee that a result is correct. Very low signal-to-noise ratios need a separate check, and the speed of sound has to be given separately. Replacing the two fractions with a likelihood ratio or a calibrated score, reporting full curves of precision and failure detection rate, and evaluating on new seeds are left for future work (Section 6.5).

### 6.3 What the ablations show

Of the six ablations, only A6 had a measurable effect; A1–A4 did not change the outcome of any scenario. Each of these four components has a design rationale (Section 3), but in the 200 mismatch-free evaluation scenarios their value was not demonstrated, and one of them, the expected-count term (A1), is the very

feature the Poisson scoring was designed around (Section 3.5). A5, which uses only `image_groups`, matches the full method and is much faster, and M6, with random starting points, fails almost every time. M6 removes both the latency hypotheses and the candidate generators, while A5 and A6 only choose between the two generators, so all that can be inferred is that the candidate generation formed by the latency hypotheses together with `image_groups` is indispensable as a whole; the contribution of the latency hypotheses alone was not isolated. Whether the remaining components help under other conditions needs separate testing, and this paper does not claim that they are necessary.

### 6.4 Why the real measurements failed

The failure of E10 lies neither in estimating the latency nor in the speed of sound: M2, given the latency, also failed every time, and the results did not change with the other speed of sound (Section 5.6). All of the analysis of causes below is exploratory and was not written down before the evaluation. The count, which does not depend on errors in the ground truth, is this: the 120 peaks span only about 19 ms, with a median of only 40 image arrivals in that range, so even if every arrival were picked, about 80 peaks would have no arrival to correspond to, far more than the 3 spurious peaks assumed by the detection model. Scored with the frozen detection model, even the M7 fits with known labels have a lower likelihood than every wrong M1 and M2 fit (Section 5.6); on these measured data the frozen scoring model prefers wrong lattices even near the ground truth, and whether the search itself also fails is a question this comparison cannot answer. The wall assumption of the detection model does not appear to be the main cause: it predicts whether an image is picked with a $\bar{\beta} = 0.8$ shared by all surfaces, but configurations in which both walls of one or two axes reflect also failed every time, and the `missed_fraction` of the ground truth was about 0.3 in every configuration (Appendix D). Nor does the peak picker appear to be the main cause. Arrival detection in measured RIRs has established methods: more than ten methods for automatically detecting the onset of an RIR have been compared, with methods based on spectra or energy preferred depending on the source type [55], and a method that takes the direct-sound segment as a template, correlates first and then refines the position and width of arrivals by dynamic time warping picked fewer false arrivals than two other methods [56]; yet with a matched filter that uses the direct sound as its template (without dynamic time warping) and with peaks of the envelope, the picked peaks in Section 5.7 still could not be told apart from random ones. The diagnosis points to the data themselves: in single responses of this room, of the first-order echoes only that of the ceiling, when the ceiling is reflective, matches the calibrated geometry to the centimetre; the floor echo is systematically about 5 cm late, possibly because the sound passes through the absorber and reflects off the hard floor beneath it; and the side-wall echoes that determine the two horizontal sides cannot be told apart from random times. The dataset authors likewise note that peaks are blurred by the loudspeaker characteristics and by several reflections arriving together [12], and that the first-order reflection from the wall directly behind a loudspeaker may be too weak to appear, so some echoes are mislabelled [12]; M7 reaches only 12.5% with the loudspeakers facing a wall and 40% with those facing the room, which is consistent with this. M7, labelled from the calibrated geometry, comes within 5 cm in only 30%, and fitting the peak positions directly in Section 5.7 gives only 31% (errors in the ground truth itself cannot be ruled out), so on these data the 5 cm success criterion is itself hard to

reach.

The simulated measurement chain does not reproduce these characteristics: the source and the microphone are ideal omnidirectional elements, and the spurious pulses of E5 are far milder than real responses. This is exactly the risk of generating and inverting data with the same model [51]: the high success rate in simulation reflects consistency within the model, not performance on real measurements. The peak pickers tried in Section 5.7 did not help; whether estimating echoes in a continuous parameter space [41], [42] can resolve these side-wall echoes was not tested in this paper. Judging from the diagnosis, lattice inversion on measurements more likely needs recordings in which the side-wall echoes are clearly resolvable, for example with an omnidirectional source, an emptier room and independently measured geometry, or responses from several positions. Under the rule frozen in advance, E10 did not adjust any parameter on dEchorate, and Section 5.7 compared peak-picker settings and clutter counts only on microphones that E10 did not use; all these changes must be evaluated on new data that have not been used for tuning.

## 6.5 Limitations

First, the main evidence comes from simulations of empty shoebox rooms with specular reflections, ideal omnidirectional transducers and white-noise excitation; the mismatch experiments are only a sensitivity analysis under mild mismatch, varying one mismatch at a time without testing combinations, and do not represent real differences. Second, the real data come from only one room, so no success rate across rooms can be computed, and the intervals of E10 describe only the sampling variation within this room, while the diagnosis of Section 5.7 also used only other microphones in the same room and the same calibrated geometry; three-dimensional lattice inversion reaches only 56% even in simulation and is still experimental. Third, the detection model, the reliability thresholds and the constants of Algorithm 1 were chosen on simulated development rooms (Appendix C), and the values tried before the choice were not recorded individually; although the evaluation uses new seeds, the scenario population is the same as in development. Fourth, the check of the dEchorate wall labels failed, so the groupings that depend on the south-wall label are unverified; the labels of M7 and the dataset's own first-order echo annotations both equal the image arrivals of the calibrated geometry (within 2 ns of each other), so there is no reference independent of the geometry, and the shares of peaks near image arrivals can only be compared with a random baseline; the uncertainty of the room side lengths is not reported, so errors in the ground truth of E10 cannot be ruled out. Fifth, the literature search relied mainly on the first 25 results of each OpenAlex query, Semantic Scholar could not be used, no Chinese-language literature was searched, and most sources were read only in their abstracts; the novelty claim covers only the sources read in full (Section 1.3). Sixth, timing was measured on one computer only: the median in a single process was 5.19 s in planar rooms and 5.27 s in three dimensions (the first 20 scenarios of E2); A5, which uses only `image_groups`, is much faster under parallel execution, but whether the planar default should switch to A5 was not tested in the evaluation. Seventh, the clutter rate and the detection profile are fixed values, and the flag uses only two fixed fractions; estimating the clutter rate and the detection profile and replacing the flag with a likelihood ratio all need to be tested on new evaluation data. The following section sums up what this paper can claim.

# 7. Conclusion

This paper used lattice inversion to test whether a shoebox room can be recovered from a single impulse response when the playback latency is unknown and the echoes are unlabelled. Within the scenario population of this paper, in mismatch-free planar simulations, it can: lattice inversion recovered the side lengths, the latency and the bearing together in 97.8% of 500 rooms, only 2.2 percentage points short of the upper bounds with a known latency and with known labels; with known labels, the time-difference least-squares method of Moore et al. [9] also succeeded every time, so what lattice inversion adds is that it needs no labels. Nearly every peak picked in these simulations was a true arrival, and the method fails when spurious peaks are numerous (0 dB), so the results of this paper cannot answer what happens when spurious peaks are numerous. Under nominal planar conditions the reliability flag marked all 11 failures, but in three dimensions 6/118 fits flagged reliable were wrong, and at 0 dB and with a wrong speed of sound it also passes wrong fits; the three-dimensional success rate was only 56%. On real measurements, with the frozen settings, every inversion failed; the flag rejected every fit, including those that succeeded with known labels, so on measured responses it did not tell right from wrong. An exploratory analysis that was not pre-specified shows that the 120 picked peaks span only about 19 ms with about 40 image arrivals in range; analyses after review also show that the frozen scoring model gives wrong lattices a higher likelihood than fits near the ground truth; whether the search itself also fails is a question this exploratory analysis cannot answer. The diagnosis of Section 5.7 further shows that in single responses of this room, of the first-order echoes only that of the ceiling, when the ceiling is reflective, matches the calibrated geometry to the centimetre, the side-wall echoes that determine the horizontal sides cannot be told apart from random times, and the peak pickers tried do not help; by this diagnosis, this failure comes mainly from the echo information a single response of this dataset carries, not only from the peak picker. The comparison of the ablations with random starting points supports only the inference that the candidate generation formed by the latency hypotheses together with `image_groups` is indispensable as a whole. Future work more likely needs measurements in which the side-wall echoes can be resolved, or responses from several positions, evaluated on new data that have not been used for tuning.

# Appendices

## Appendix A. Candidate generation: steps and constants

**Algorithm 1.** Candidate generation (run once for each latency hypothesis $\delta$; $q_j = r_j - \delta$, $q_0$ is the direct sound, $\epsilon = \max(1 \text{ sample}, 5 \text{ mm})$)

1. Latency hypotheses: each pair $(A, B)$ among the first 11 reflection echoes, combined with each later echo as $A+B$, gives $\delta$ by the formula of Section 3.2; combinations with a denominator below 0.2 m, or with $\delta$ outside $(-5 \text{ m},\ r_0 - 0.2 \text{ m})$, are discarded. The score of each $\delta$ is the number of early echo pairs with an echo within 1.5 cm of the predicted position of their combination; the top 8 by score that are at least 8 cm apart are kept.
2. Identity graph: echoes $a$ and $b$ (neither of them the direct sound) are joined if an echo lies within

$\epsilon$ of $\sqrt{q_a^2 + q_b^2 - q_0^2}$.

3. Seeds: among the first 14 reflection echoes, a joined pair in planar rooms; in three dimensions, three echoes with all three pairs joined, or with two pairs joined and an echo within $\epsilon$ of $\sqrt{q_a^2 + q_b^2 + q_c^2 - 2q_0^2}$. Seeds are ranked by the number of joins and by total length, and at most 6 are kept.
4. Grouping: the group of axis $i$ is the set of echoes joined to the seeds of all the other axes; echoes joined to a fraction below 0.3 of the members of the other groups are removed (seeds are kept).
5. Single-axis fit: $c_i$ on a grid from 0 to $q_0$ with 5 mm spacing; the shortest member is taken as one first-order wall and paired with one of the 2nd- to 4th-shortest members, which is read as the other first-order wall or as the image at $2L_i - c_i$ or $2L_i + c_i$, with $L_i > 0.5$ m and the source and microphone inside the walls. The score is the number of group members explained by the first 10 single-axis images of this axis, plus 0.25 times the number of other echoes explained (tolerance $2\epsilon$); the top 3 that differ by more than 3 cm in side length or by more than 5 cm in $c_i$ are kept.
6. Closure condition: when all the other axes have fits, the axis that lacks a second-order reflection takes $c_i^2 = q_0^2 - \sum_{j \neq i} c_j^2$, with its walls determined by the two shortest members of its group.
7. Combination: each combination of the per-axis fits becomes a candidate room if $|\|\mathbf{c}\| - q_0| \leq 0.1$ m.
8. Scoring and refinement: all candidates are ranked by the likelihood of Section 3.5 (kernel width $\max(2 \text{ cm}, 1 \text{ sample})$), and candidates whose largest parameter difference is below 5 cm are treated as duplicates; the best 8 (12 in three dimensions) go on to the fitting of Section 3.6, and the best 2 of these (3 in three dimensions) are also rescanned axis by axis.

## Appendix B. Overview of the experiments

**Table B1.** Overview of the experiments. "Exploratory" means an analysis that was not written down before the evaluation; those numbered PR1–PR13 were added after review. For the peak-picking diagnosis and for PR1–PR13, the last column ends with the sections where they are reported.

| Label | Sub-question | Scenarios or responses | Methods compared | Status |
|---|---|---|---|---|
| E1 | Planar, no mismatch | 500 | M1–M8 (M6 on the first 200) | Pre-specified (M8 added after review) |
| E1b, E1t | Scope of the first-order closed-form solver; timing | 100, 20 | M1, M4, M5; M1–M3 | Pre-specified |
| E2 | Three-dimensional, no mismatch | 200 | M1, M2, M6 (first 100), M7 | Pre-specified |
| E3, E4 | Signal-to-noise ratio; latency | 100 per level | M1, M5; M1, M2 | Pre-specified |
| E5, E9 | Model mismatch; distribution shift | 100 per condition | M1 | Pre-specified |
| E6 | Ablations A1–A6 | 200 (first 200 of E1) | M1 and six variants | Pre-specified |

| Label | Sub-question | Scenarios or responses | Methods compared | Status |
|---|---|---|---|---|
| E7, E8 | Reliability flag; standard errors | E7: 1,700 fits of E1, E3 and E5; E8: 489 successful fits of E1 (three dimensions: see E2) | M1 | Pre-specified |
| E10 | dEchorate measurements | 540 (plus 54 with furniture) | M1, M2, M7 | Pre-specified |
| Peak-picking dia-gnosis | Span of the picked peaks; share of peaks near image arrivals | E10 54, E2 20 | — | Exploratory, not pre-specified; §5.6 |
| PR1 | Method of Moore et al. [9] (M8) | E1 500 | M8 with M1, M2, M7 | Added after review; §4.2, §5.1 |
| PR2 | Failed scenarios of E1, rerun with the same seed, source of the M1 row of Table 4 | E1 500 | M1 | Added after review; §5.1, §5.4 |
| PR3 | Spurious and missed peaks per scenario | E1 500, E2 200; R4, 10 dB, 0 dB 100 each | True lattice | Added after review; §1.3 |
| PR4 | Share of peaks near image arrivals at three tolerances, with two random baselines | E10 54, E2 20 | True lattice | Added after review; §5.6, Fig. 3 |
| PR5 | Likelihood under the frozen model; results by wall configuration | E10 540 | Ground truth, M1, M2, M7 | Added after review; §5.6, Appendix D |
| PR6 | Random baseline for `explained_fraction` | E1, 10 dB, 0 dB, E10 | M1 | Added after review; §6.2 |
| PR7 | Coverage conditioned on being flagged reliable | E1 489, E2 118 | M1 | Added after review; §5.5 |
| PR8 | Per-condition 2 × 2 tables of E7, clustered bootstrap, third cell; source-clustered interval for E10 | E7 1,700; E10 540 | M1, M7 | Added after review; §5.5, §5.6 |
| PR9 | Single-process timing in three dimensions | First 20 of E2 | M1 | Added after review; §6.5 |
| PR10 | Peak-to-noise ratio of the identified responses | E3 500 | — | Added after review; §4.1, §5.3 |
| PR11 | Onset check by source type | E10 540 | — | Added after review; §5.6 |

| Label | Sub-question | Scenarios or responses | Methods compared | Status |
|---|---|---|---|---|
| PR12 | First-order echoes: dataset annotations against the calibrated geometry | 360 E10 responses from the directional loudspeakers | Annotations, calibrated geometry | Added after review; §5.6, §6.5 |
| PR13 | Diagnosis of the measured failure: first-order echo times, peak pickers and end-to-end inversion | Microphones not used in E10: 360 for development, 360 for confirmation | Calibrated geometry, three peak pickers (37 settings), M1 | Added after review, criteria written down before the responses used were read; §5.7, Appendix E |

## Appendix C. Development stage

The development stage used a separate set of scenarios: 40 planar and 20 three-dimensional scenarios generated from the same scenario population with master seed 0, and 24 simulated measurements in 8 planar rooms (latency 4 ms), none of which were reused in the evaluation. The planar reliability thresholds were set from two wrong fits during development, whose `explained_fraction` and `missed_fraction` were 0.875 and 0.234, and 0.825 and 0.146; the three-dimensional thresholds, the parameters of the detection model ($\bar{\beta} = 0.8$, $n_c = 3$, a picking threshold of 2%) and the constants of Algorithm 1 were also chosen during development, and the other values tried before the choice were not recorded individually. On the development set the success rate of M1 was 39/40 (planar) and 11/20 (three-dimensional); all fits flagged reliable succeeded, and all failures were flagged unreliable (1/1 planar, 9/9 three-dimensional).

## Appendix D. E10 results by wall configuration

**Table D1.** E10 results by wall configuration (exploratory analysis after review). The codes give, in order, the floor, ceiling, west, south, east and north surfaces, with 1 for reflecting; "axes with both walls reflecting" counts only the horizontal axes, since the floor always absorbs. The last two columns are the two fractions of the frozen detection model on the calibrated ground truth (medians). 54 responses per configuration.

| Configuration | Axes with both walls reflecting | M1 successes | M2 successes | M7 success rate | Ground-truth explained | Ground-truth missed |
|---|---|---|---|---|---|---|
| 000000 | 0 | 0/54 | 0/54 | 29.6% | 0.208 | 0.322 |
| 000001 | 0 | 0/54 | 0/54 | 33.3% | 0.217 | 0.328 |
| 000010 | 0 | 0/54 | 0/54 | 35.2% | 0.225 | 0.337 |
| 000100 | 0 | 0/54 | 0/54 | 37.0% | 0.208 | 0.342 |
| 001000 | 0 | 0/54 | 0/54 | 22.2% | 0.212 | 0.330 |
| 010000 | 0 | 0/54 | 0/54 | 31.5% | 0.221 | 0.337 |
| 011000 | 0 | 0/54 | 0/54 | 27.8% | 0.208 | 0.320 |
| 011100 | 0 | 0/54 | 0/54 | 29.6% | 0.225 | 0.309 |

| Configuration | Axes with both walls reflecting | M1 successes | M2 successes | M7 success rate | Ground-truth explained | Ground-truth missed |
|---|---|---|---|---|---|---|
| 011110 | 1 | 0/54 | 0/54 | 27.8% | 0.221 | 0.294 |
| 011111 | 2 | 0/54 | 0/54 | 25.9% | 0.225 | 0.311 |

## Appendix E. Diagnosis of the measured failure

The diagnosis (PR13) used microphones that E10 did not use: capsule 1 of each array for development (360 responses) and capsule 5 for confirmation (360 responses), covering the 10 empty-room wall configurations and the 6 directional loudspeakers; the middle capsule of each array, which E10 used, was not read. The criteria were written down before these responses were read (Table E2), and the peak-picking settings were compared on the development set only; the confirmation set only confirms the procedure chosen on the development set, and since the peak-picking and end-to-end criteria had already failed there, only the comparison of first-order echo times and the fit with labels given were repeated on the confirmation set. Predicted and measured first-order echoes are both timed relative to the direct sound, so a common transducer delay does not affect the offsets. On the microphones of E10, none of the peaks that the picker took as the direct sound of the two loudspeakers facing a wall lay within one sample of the calibrated direct sound (Section 5.6), so Table E1 and the end-to-end inversion use only the loudspeakers facing the room, while the fit with labels given and the comparison of peak pickers use all 360 development responses. Counting only the 240 facing-room responses, the fit with labels given came within 5 cm and 10 cm in 34% and 82.5% on the development set and in 26% and 79.6% on the confirmation set, close to the results for all 360.

Compared with the plan written in advance, the procedure differs in three ways that affect the results here. First, the plan searched for peaks within ±0.5 ms of each prediction, and ±0.25 ms was used; a narrower range can only bring the peaks found closer to the prediction, so the fit with labels given leans optimistic. Second, the plan's end-to-end inversion was to run M1 and M2, with the latency known, and to try a version that estimates the clutter count for each response, on all development responses; only M1 was run, only on the 240 facing-room responses, with 4 picker combinations, one of them with the clutter count fixed at 60, and neither M2 nor the per-response clutter estimate was run. Third, the plan allowed the reliability threshold to be recalibrated on the development set, but with at most 1 of the 240 responses within 10 cm no threshold can meet the end-to-end criterion, so it was not recalibrated.

**Table E1.** Measured minus predicted first-order echoes, as path length, for the 4 loudspeakers facing the room in the development set (240 responses). The measured position is the maximum of the absolute response within ±0.25 ms of the prediction; "random" repeats the same search at uniformly drawn times; amplitudes are medians.

| First-order echo | Count | Median offset | 1st to 3rd quartile | Within 2 cm | Within 2 cm at random | Amplitude (relative to direct) |
|---|---|---|---|---|---|---|
| Ceiling (reflecting) | 120 | −0.4 cm | −1.4 to +1.3 cm | 67% | 14% | 0.45 |
| Ceiling (absorbing) | 120 | +2.3 cm | −3.1 to +5.6 cm | 17% | 23% | 0.13 |
| Floor (always absorbing) | 240 | +5.3 cm | +3.5 to +6.1 cm | 10% | 19% | 0.29 |
| Four side walls (reflecting) | 336 | −1.0 cm | −5.1 to +5.2 cm | 14% | 17% | 0.14 |
| Four side walls (absorbing) | 624 | +0.8 cm | −4.0 to +5.6 cm | 20% | 21% | 0.10 |

**Table E2.** Criteria and results. The peak pickers were 37 settings: the frozen picker, and a matched filter with the direct sound as its template and the response envelope, each of the latter two with peak spacings of 6, 12 and 24 samples and thresholds of 2%, 5% and 10% of the direct sound, with and without skipping the first 1 ms after the direct sound; the ratio is the median over the 360 development responses of each setting's per-response ratio, and the recall is the share of first-order echoes from reflecting walls with a peak within 2 cm (mean over responses). End to end, M1 was run with 4 of these combinations (one of them with a clutter count of 60).

| Criterion | Development set | Confirmation set |
|---|---|---|
| Data: a fit to the measured peak positions with the labels given comes within 5 cm in at least 80%; otherwise 10 cm is the bar, and if not even 10 cm reaches 80% the diagnosis stops | 31% within 5 cm, 80.0% within 10 cm (288/360) | 28% within 5 cm, 79.7% within 10 cm |
| Peak picking: with the best picker, the share of picked peaks within 2 cm of an image arrival is at least 2 times the random baseline, and the recall of first-order echoes from reflecting walls is at least 80% | Ratio 0.85–1.02, recall 20%–65% | Not tested again (failed on the development set) |
| End to end: M1 succeeds in at least 30% at the threshold set by the data criterion (10 cm); at least 80% of the fits flagged reliable are correct, with at least 20% of the responses flagged (240 facing-room responses) | 0% within 5 cm, at most 0.4% within 10 cm (1/240), no fit flagged reliable | Not tested again |

| Criterion | Development set | Confirmation set |
|---|---|---|
| Confirmation: the same procedure on the confirmation set, with the success rate and the precision of reliable fits no more than 10 percentage points below the development set | — | Not run (failed on the development set) |

# Declarations

## Data and code availability

The code of lattice inversion will be released under the MIT licence at https://github.com/recklight/echo-lattice; the experiments in this paper used the code of commit `ffe8311`, and the commit `1948ea0` recorded in the freeze file and the result files is the same code (git tree `b12d5c4`). In the same repository, `paper/` will hold the Chinese and English PDFs of this paper (under CC BY 4.0), and `experiments/` will hold the scripts of every experiment, of the analyses added after review and of the diagnosis in Section 5.7, together with the copy of the freeze file. The experimental plan and settings were frozen before the evaluation, but the time of freezing is recorded only in the author's internal records and cannot be verified independently by readers. The per-scenario result tables (including PR1–PR13, added after review, and the per-condition 2 × 2 tables) are available from the author on request. The freeze file provided is a copy: the copy changes only the source URL of the lattice inversion code in the package list to the current repository name and is otherwise byte-identical to the original, so the two hashes differ. The SHA-256 of the original is `92b6696dd35f9b9144aaf61211fdac6caf15ed635139d02d1b6df73ffc492430`, and that of the copy is `56ba65e43c393fb7649dd640a83ba967466d6203012db34bd49066e800f27663`. The real measurements are taken from the SOFA version of the dEchorate dataset [12] (https://sofacoustics.org/data/database/dechorate/); the data record of the SONICOM Ecosystem gives its licence as Creative Commons Attribution 4.0 (CC BY 4.0), while the licence attribute inside the files is the MIT licence, and this paper credits the source under both.

## Ethics

This study involved no human or animal subjects. The real measurements use only room impulse responses measured with swept-sine excitation in the public dEchorate dataset.

## Author contributions

Following the roles of the Contributor Roles Taxonomy (CRediT), Fu-Kai Chuang was responsible for conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing of the original draft, and review and editing.

### Conflicts of interest

The author declares no conflicts of interest.


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